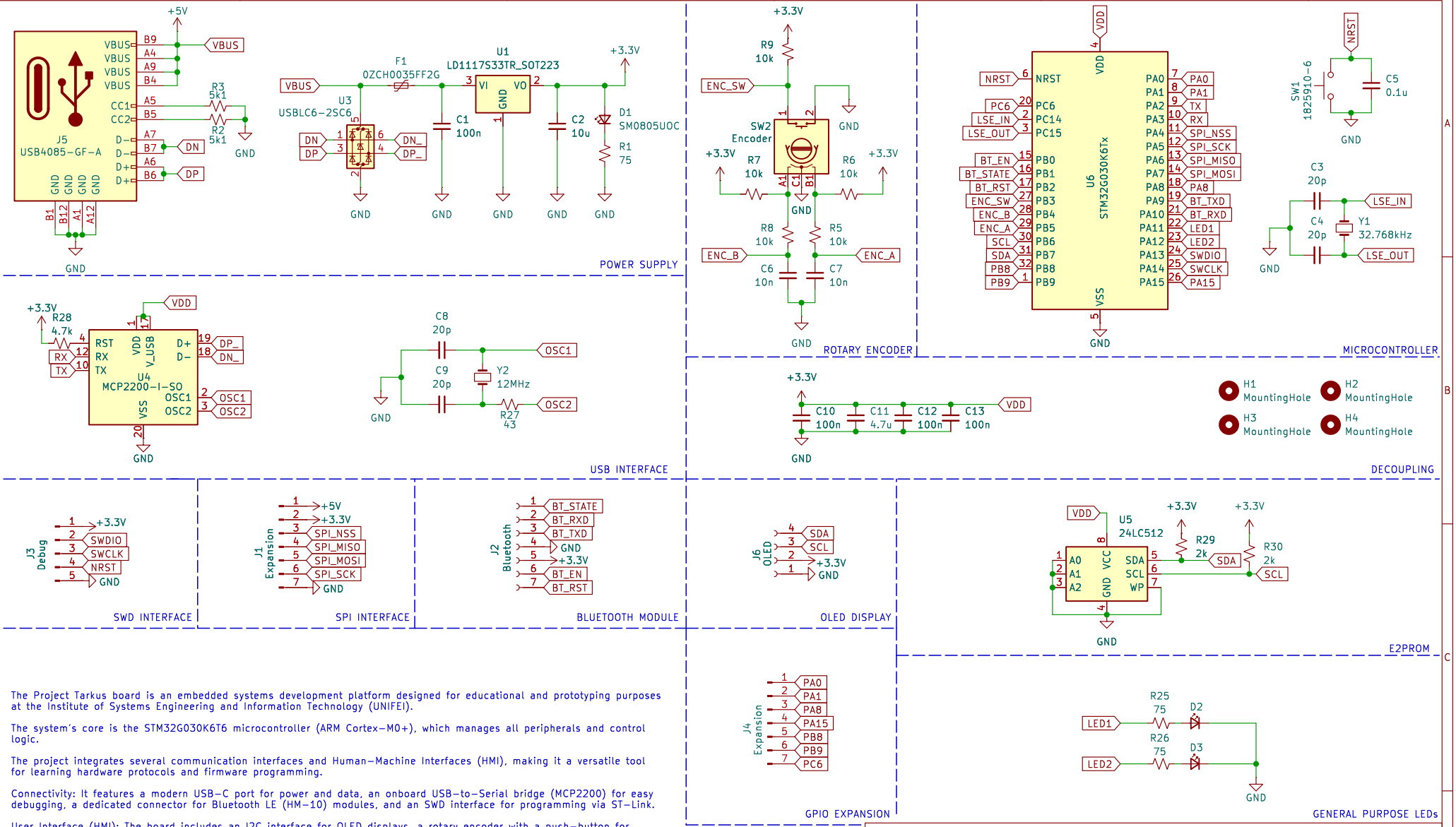

The Project Tarkus board is an embedded systems development platform designed for educational and prototyping purposes at the Institute of Systems Engineering and Information Technology (UNIFEI).

The system's core is the STM32G030K6T6 microcontroller (ARM Cortex-M0+), which manages all peripherals and control logic.

The project integrates several communication interfaces and Human-Machine Interfaces (HMI), making it a versatile tool for learning hardware protocols and firmware programming.

Connectivity: It features a modern USB-C port for power and data, an onboard USB-to-Serial bridge (MCP2200) for easy debugging, a dedicated connector for Bluetooth LE (HM-10) modules, and an SWD interface for programming via ST-Link.

User Interface (HMI): The board includes an I2C interface for OLED displays, a rotary encoder with a push-button for menu navigation, and two general-purpose LEDs for status indication.

Storage & Timing: It is equipped with an external EEPROM (24LC512) for non-volatile data storage and a 32.768kHz crystal to ensure high precision for the internal Real-Time Clock (RTC).

Power & Expansion: The system is powered via USB with 3.3V regulation and surge protection. Unused pins (including SPI and ADC lines) are broken out to an expansion header, allowing for the connection of external sensors and modules.

Designed by Bernardo Lamim Fogaça de Padua
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